

SQL :

create database bank;

use bank;

create table BR (^{brn} varchar(20) primary key, ct varchar(20),
asst int);insert into BR values ('BR-a', 'Bk', 50000), ('BR-b',
'delhi', 30000), ('BR-c', 'Mumbai', 40000);

select * from BR;

brn	ct	asst
BR-a	Bk	50000
BR-b	delhi	30000
BR-c	Mumbai	40000

create table AC (ano int primary key, brn varchar(20),
bal int, foreign key (brn) references BR (brn));insert into AC values (101, 'BR-a', 5000), (102, 'BR-a',
7000), (103, 'BR-b', 3000), (104, 'BR-c', 9000), (105,
'BR-b', 4000);

select * from AC;

ano	brn	bal
101	BR-a	5000
102	BR-a	7000
103	BR-b	3000
104	BR-c	9000
105	BR-b	4000

create table CM (comn varchar(20) primary key, ct varchar(20),

insert into CM values ('C1', 'Bk'), ('C2', 'delhi'), ('C3',
'Mumbai'), ('C4', 'delhi');

select * from CM

cmn	ct
C1	Bh
C2	Delhi
C3	Mumbai
C4	Delhi

```
select * from AC where bal > 5000;
```

ano	lcn	bal
102	BR-a	7000
104	BR-c	5000

```
select * from CM where ct = 'Delhi';
```

cmn	ct
C2	Delhi
C4	Delhi

```
select * from AC order by bal desc;
```

ano	lcn	bal
104	BR-c	9000
102	BR-a	7000
101	BR-a	5000
105	BR-b	4000
103	BR-b	3000

MongoDB shell:

```
> use college
college >
```

```
> db.student.insert one({
  name: 'Alice',
  age: 20,
  dept: 'CSE'
})
```

```
> db.student.insert many ( {  
  { name: 'Bob', age: 21, dept: 'Mech' },  
  { name: 'Aria', age: 22, dept: 'CSE' } )
```

```
> db.student.find ( )
```

```
> db.student.find ( { dept: 'CSE' } )
```

```
> db.student.find ( { age: { $gt: 20 } } )
```

```
> db.student.updateOne (   
  { name: 'Alice' },  
  { $set: { age: 21 } }  
 )
```

```
> db.student.deleteOne ( { name: 'Bob' } )
```

```
> exit
```

```
> _MONGOSH
```

```
> db["Employee_Info"].find()
```

```
<
```

```
> use Employee
```

```
< switched to db Employee
```

```
> db.createCollection("Employee_Info")
```

```
< { ok: 1 }
```

```
> db.Employee_Info.createIndex({ Emp_Id: 1 }, { unique: true })
```

```
❶ • MongoServerError[DuplicateKey]: Index build failed: 5778370
```

```
> db.Employee_Info.insertMany([
```

```
{
```

```
  _id: 121,
```

```
  Emp_Name: "Alice",
```

```
  Designation: "Software Engineer",
```

```
  Date_of_Joining: ISODate("2021-06-15"),
```

```
  Salary: 75000,
```

```
  Dept_Name: "IT"
```

```
},
```

```
{
```

```
  _id: 122,
```

```
  Emp_Name: "Bob",
```

```
  Designation: "Senior Engineer",
```

```
  Date_of_Joining: ISODate("2020-03-10"),
```

```
  Salary: 90000,
```

```
  Dept_Name: "R&D"
```

```
},
```

```
{
```

```
  _id: 123,
```

```
  Emp_Name: "Charlie",
```

```
  Designation: "Manager",
```

```
  Date_of_Joining: ISODate("2019-01-20"),
```

```
  Salary: 105000,
```

```
  Dept_Name: "HR"
```

```
}]
```

```
})
```

```

< {
  acknowledged: true,
  insertedIds: {
    '0': 121,
    '1': 122,
    '2': 123
  }
}
> db.Employee_Info.updateOne(
  { _id: 121 },
  {
    $set: {
      Emp_Name: "Alice Johnson",
      Dept_Name: "Engineering"
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.Employee_Info.find().sort({ Salary: 1 })
< {
  _id: 121,
  Emp_Name: 'Alice Johnson',
  Designation: 'Software Engineer',
  Date_of_Joining: 2021-06-15T00:00:00.000Z,
  Salary: 75000,
  Dept_Name: 'Engineering'
}
{
  _id: 122,
  Emp_Name: 'Bob',
  Designation: 'Senior Engineer',

```

```
{
  _id: 122,
  Emp_Name: 'Bob',
  Designation: 'Senior Engineer',
  Date_of_Joining: 2020-03-10T00:00:00.000Z,
  Salary: 90000,
  Dept_Name: 'R&D'
}
{
  _id: 123,
  Emp_Name: 'Charlie',
  Designation: 'Manager',
  Date_of_Joining: 2019-01-20T00:00:00.000Z,
  Salary: 105000,
  Dept_Name: 'HR'
}
> db.Employee_Info.find().sort({ Salary: -1 })
< {
  _id: 123,
  Emp_Name: 'Charlie',
  Designation: 'Manager',
  Date_of_Joining: 2019-01-20T00:00:00.000Z,
  Salary: 105000,
  Dept_Name: 'HR'
}
{
  _id: 122,
  Emp_Name: 'Bob',
  Designation: 'Senior Engineer',
  Date_of_Joining: 2020-03-10T00:00:00.000Z,
  Salary: 90000,
  Dept_Name: 'R&D'
}
{
  _id: 121,
  Emp_Name: 'Alice Johnson',
  Designation: 'Software Engineer',
```

```

{
  _id: 121,
  Emp_Name: 'Alice Johnson',
  Designation: 'Software Engineer',
  Date_of_Joining: 2021-06-15T00:00:00.000Z,
  Salary: 75000,
  Dept_Name: 'Engineering'
}
> db.Employee_Info.updateOne(
  { _id: 121 },
  {
    $addToSet: {
      Projects: { $each: ["Payroll System", "Cloud Migration"] }
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.Employee_Info.updateMany(
  {},
  {
    $set: {
      expireAt: new Date(Date.now() + 15000)
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}

```

```

> db.Employee_Info.updateMany(
  {},
  {
    $set: {
      expireAt: new Date(Date.now() + 15000)
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
> db.Employee_Info.updateMany(
  {},
  {
    $set: {
      expireAt: new Date(Date.now() + 15000)
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
> db.Employee_Info.createIndex(
  { expireAt: 1 },
  { expireAfterSeconds: 0 }
)
< expireAt_1
>
Employee>

```